$$F_{ext} = BI\left(\frac{\quad}{3}\right) = B\left(\frac{\quad}{r}\right)\left(\frac{\quad}{3}\right) = \frac{\quad}{3r}$$

26 A

$$\varepsilon_{ind} = \frac{\Delta BA}{t} = \frac{\left(B\sin 30^{o}\right)A}{t}$$

$$i = \frac{\varepsilon_{ind}}{R}$$

$$Q = it = \frac{\varepsilon_{ind}}{R}t = \frac{\left(B\sin 30^{o}\right)A}{R}$$

$$= \frac{\left(0.70\times\sin 30^{o}\right)\left(16.0\times10^{-4}\right)}{\left(2.0\times10^{-3}\right)}$$

$$= 0.28\ C$$

27 B $hf_i = hf_f + KE$